PROJECT TITLE:

SMARTCDS

WHERE EFFICIENCY MEETS APPETITE,
SERVING UP BILLING BRILLIANCE!"

Meet The Team

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PROJECT OVERVIEW

SmartCDS is a software project aimed at enhancing the efficiency and accuracy of billing processes within our university's canteen. The primary goal is to create a user-friendly and comprehensive system that aims to **eliminate** the overdue payment from customers. The application will be developed in C++ to ensure performance and flexibility.

Problem Statement



Issue 1

Records are **scrambled** and not stored, causing data disarray and loss of vital information

Issue 2

The university canteen's **billing system** is **inefficien**t and error-prone due to manual procedures, leading to inaccuracy

Issue 3

Lacks **finanacial transperency** as there's no effective sales tracking or streamlined credit sales management.

OUR MOTIVATION TO MAKE THIS SYSTEM



CREDIT SALES MANAGEMENT



SALES TRACKING



ACCURACY IMPROVEMENT

Features in the System

Order Placement

Product Management

Administration

Discount Calculation

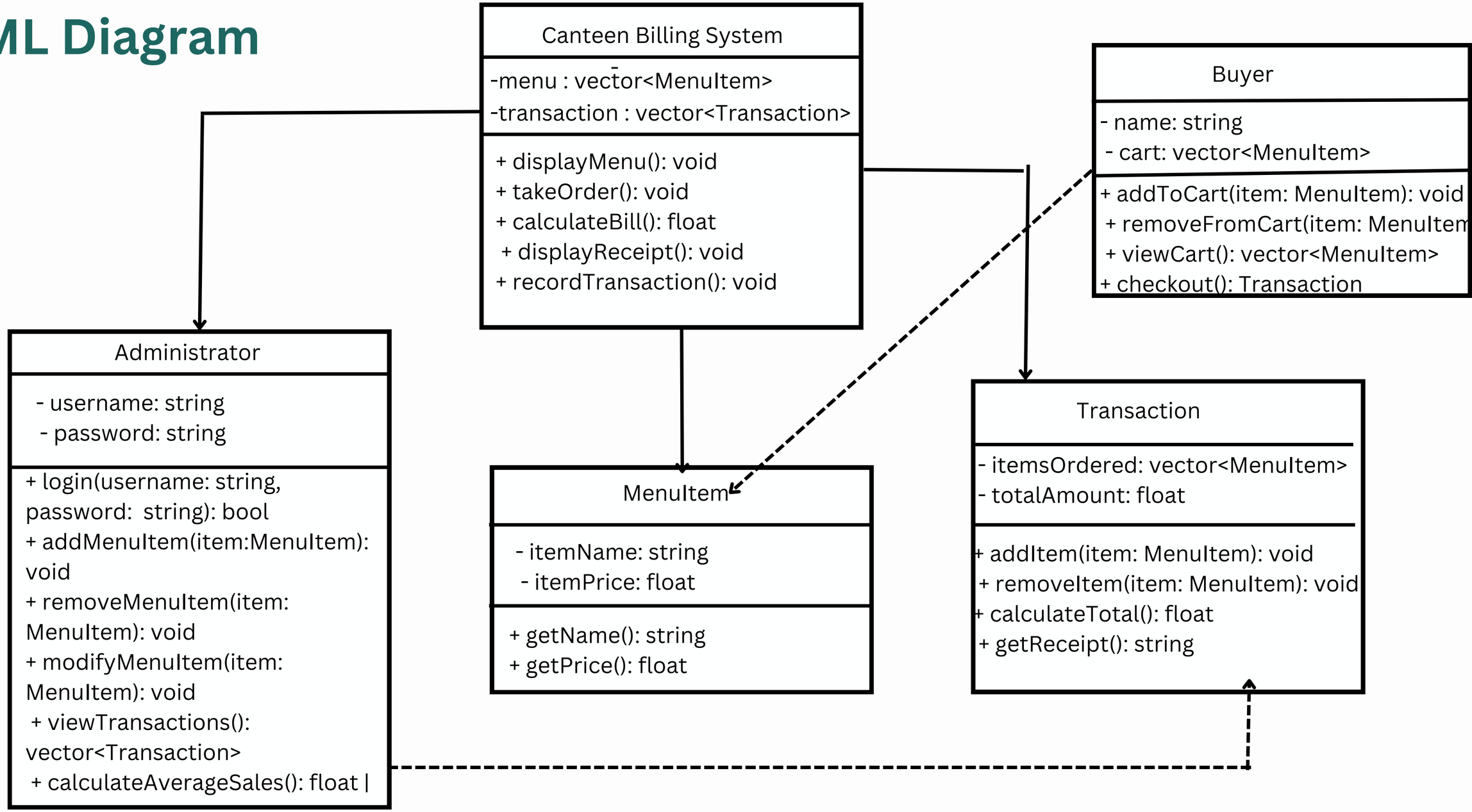
Buyers

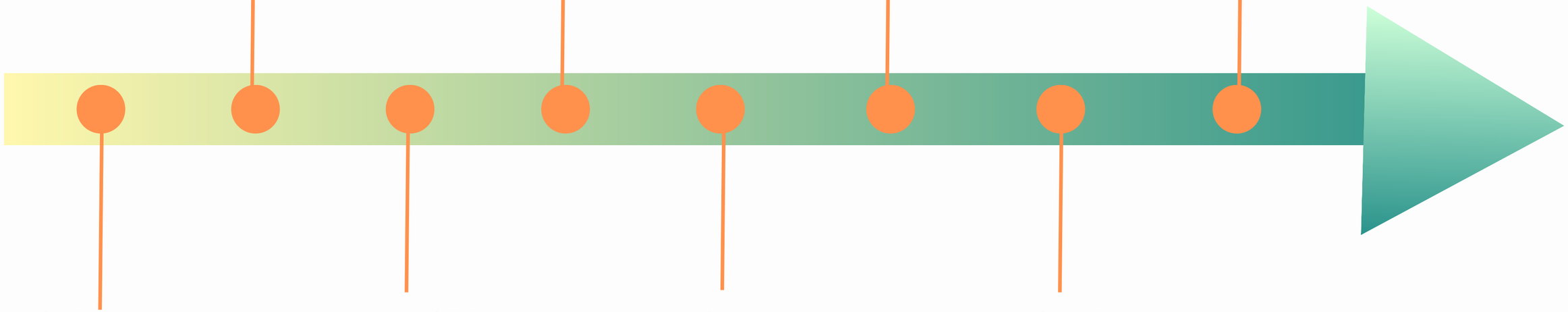
Record Modification

Receipt Generation

Credit Sales Management

UML Diagram


Canteen Billing System
-menu : vector<MenuItem>
-transaction : vector<Transaction>
+ displayMenu(): void
+ takeOrder(): void
+ calculateBill(): float
+ displayReceipt(): void
+ recordTransaction(): void
Buyer
- name: string
- cart: vector<MenuItem>
+ addToCart(item: MenuItem): void
+ removeFromCart(item: MenuItem
+ viewCart(): vector<MenuItem>
+ checkout(): Transaction
Administrator
- username: string
- password: string
+ login(username: string, password: string): bool
+ addMenuItem(item:MenuItem): void
+ removeMenuItem(item: MenuItem): void
+ modifyMenuItem(item: MenuItem): void
+ viewTransactions(): vector<Transaction>
+ calculateAverageSales(): float |
MenuItem
- itemName: string
- itemPrice: float
+ getName(): string
+ getPrice(): float
Transaction
- itemsOrdered: vector<MenuItem>
- totalAmount: float
+ addItem(item: MenuItem): void
+ removeItem(item: MenuItem): void
+ calculateTotal(): float
+ getReceipt(): string



Week 1-2:
Project Initiation and Planning
• Define project scope, objectives, and deliverables.
• Create a detailed project plan.
• Set up project infrastructure (development environment, version control, etc.).

Week 3-4:
Requirements Gathering and Analysis
• Gather detailed requirements for the billing system.
• Create a requirements specification document, including a timeline, budget and resource allocation.

Proposal Presentation.

Week 5-6
: System Design
• Design a high-level system architecture.
• Design the structure of text files for storing customer information, invoices and billing data.

Week 9-11:
Development
• Start coding the core functionalities.
• Implement User authentication and access control (using text files).
• Implement product management features for administrators.
• Develop order placement and receipt generation features for buyers.
• Develop the core billing logic, including invoice generation and payment

Week 12:
Progress Evaluation
• Evaluate project progress against the timeline.
• Ensure alignment with project objectives.
• Make adjustments as necessary.

Week 13-14:
Development and Testing
• Continue coding and refining features.
• Perform code reviews and address issues.
• Implement discount calculation and menu management features.
• Complete remaining development tasks.

Week-15:
Final Presentation

Progress Timeline

	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-11	Week 12	Week 13-14	Week 15
Project Initiation and Planning								
Requirements Gathering and Analysis								
System Design								
Proposal Presentation								
Development								
Progress Evaluation								
Development and Testing								
Final Presentation								

Completed

Not finished yet

Ongoing